Computer Networks and Internets with Internet Applications, 4e

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Chapter 34

File Transfer And Remote File Access

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34.1 Introduction

This chapter

- presents a network application that can transfer copies of a file from one computer to another
- discusses the file transfer interface
- considers file access and explains how the underlying software uses the client-server paradigm

34.2 Generalized File Transfer

A SW system that sends a copy of arbitrary data from a file on one computer to a file on another

- is known as "file transfer" SW

- File transfer is complicated because an internet can connect heterogeneous computer systems
  - File transfer SW must accommodate differences among the ways computer systems store files
  - The owner on one computer system may not have a corresponding login account on another system
  - File transfer SW must accommodate other minor differences in file representations, type information, and file protection mechanisms

34.3 The File Transfer Protocol (1)

- The most widely-deployed Internet file transfer service uses the
  - File Transfer Protocol (FTP)
- The characteristics of FTP are:
- General-Purpose
  - handles many of the problems discussed above.
- Arbitrary File Contents
  - permits transfer of arbitrary data
- Authentication And Ownership
  - includes a mechanism that allows files to have ownership and access restrictions
- Accommodates Heterogeneity
  - hides the details of individual computer systems
  - it can be used to transfer between an arbitrary pair of computers

34.3 The File Transfer Protocol (2)

- FTP is among the oldest application protocols still used in the Internet
  - It is invoked by browsers when a user requests a file download
- Originally defined as part of the ARPANET protocols
  - FTP predates both TCP and IP
- As TCP/IP was created, a new version of FTP was developed that worked with the new Internet protocols
- FTP is heavily used
  - Early in the history of the Internet
    - FTP accounted for approximately one-third of all Internet traffic
    - the traffic generated by services such as e-mail and the DNS did not come close to exceeding that generated by FTP

34.4 FTP General Model And User Interface

- FTP is designed to be run from a program (e.g., from a browser) or for interactive use
- When a program invokes FTP
  - the program must handle all details and then inform the user whether the operation succeeded or failed; the user never sees the FTP interface
- When a user invokes FTP interactively
  - the user communicates with a command-driven interface
- FTP issues a prompt to which the user responds by entering a command
- FTP executes the command, and then issues another prompt
- FTP has commands that
  - allow a user to connect to a remote computer
  - provide authorization
  - find out which remote files are available
  - and request file transfer of one or more files
- Some FTP commands require little or no time to execute, while others can take a significant time

34.5 FTP Commands (1)

- Although the FTP protocol standard specifies exactly how FTP SW on one computer interacts with FTP software on another
  - the standard does not specify a user interface
- Consequently, the interface available to a user can vary from one implementation of FTP to another
- To help maintain similarity among products
  - many vendors have chosen to adopt the interface that first appeared in an early version of FTP software written for the BSD UNIX system
- The BSD interface for FTP supports over 50 individual commands
- Figure 34.1 lists the command names

!	cr	macdef	proxy	sendport
$	delete	mdelete	put	status
account	debug	mdir	pwd	struct
append	dir	mget	quit	sunique
ascii	disconnect	mkdir	quote	tenex
bell	form	mls	recv	trace
binary	get	mode	remotehelp	type
bye	glob	mput	rename	user
case	hash	nmap	reset	verbose
cd	help	ntrans	rmdir	?
cdup	lcd	open	runique	
close	ls	prompt	send	

Figure 34.1 The names of commands found in the BSD interface for FTP. Many vendors support a variant of the BSD interface.

34.5 FTP Commands (2)

- The list of commands can seem overwhelming to a beginner because
  - BSD interface contains options that are seldom implemented
    - (e.g., the proxy command that permits simultaneous communication with two remote sites)
  - The interface provides several options to handle obscure details, some of which have become irrelevant
    - For example, tenex refers to an obsolete OS
  - FTP also accommodates multiple representations of texts
  - A final form of complexity arises because the BSD interface also includes aliases (i.e., multiple names)

34.6 Connections, Authorization, And File Permissions

- Fortunately, only a handful of FTP commands are used
  - After starting an FTP program
    - a user must enter the "open" before any files can be transferred
  - Open requires
    - the user to give the domain name of a remote computer
    - and then forms a TCP connection, known as a "control connection"
    - A TCP connection to a remote machine is used to send commands
  - FTP requests the user to supply authorization
    - the user must enter a login name and a password
  - After a user opens a control connection → obtains authorization → the user can transfer files
  - When a user finishes accessing a particular computer, the user enters the "close" to terminate connection

34.7 Anonymous File Access

- To permit arbitrary users to access a file
  - many sites follow the convention of establishing a special computer account used only for FTP
- The account, which has the login name "anonymous"
  - permits an arbitrary user minimal access to files
  - early systems used the password "guest" for anonymous access
- More recent versions of FTP often request that
  - a user send their e-mail address as a password
  - making it possible for the remote FTP program to send the user e-mail if problems arise
- The term "anonymous FTP" is used
  - to describe the process of obtaining access with the anonymous login name

34.8 File Transfer In Either Direction

- Interestingly, FTP allows file transfer in either direction
- After a user establishes a connection to a remote server
  - the user can obtain a copy of a remote file
    - get or mget
  - or transfer a copy of a local file to the remote machine
    - put or mput
- Such transfers are subject to access permissions
  - the remote computer can be configured to prohibit creation of new files or changes to existing files
  - and the local computer enforces conventional access restrictions on each user

34.9 Wildcard Expansion In File Names

- To make it easy for users to specify a set of file names
- FTP allows a remote computer system to perform traditional file name expansion
- The user enters an abbreviation, which FTP expands to produce a valid file name.
- In abbreviations, a “wildcard” character stands for zero or more characters
  - Many computer systems use the asterisk (*) as a wildcard

34.10 File Name Translation

- FTP can be used between heterogeneous computer systems
  - the SW must accommodate differences in file name syntax
    - some systems restrict file names to uppercase letters
    - while others permit a mixture of lowercase and uppercase.
    - some systems permit a file name to contain as many as 128 characters, while others restrict names to 8 or fewer characters
- File name differences can be especially important when using an abbreviation (e.g., on an mget or mput command)
  - a user can specify an abbreviation that FTP expands into a list of names
  - a file name that is valid on one computer may be illegal on another
- To handle incompatibilities among computer systems
  - the BSD interface to FTP permits a user to define rules that specify how to translate a file name when moving to a new computer system
    - Thus, a user can specify that FTP should translate each lowercase letter to its uppercase equivalent

34.11 Changing Directories And Listing Contents

- Many computers have a hierarchical file system that places each file in a “directory” some systems use the term “folder”
- The hierarchy arises because a directory can contain other directories as well as files
- FTP supports a hierarchical file system by including the concept of a current directory
  - all file names are interpreted in the current directory
  - and all file transfers affect the current directory
- Command “pwd” can be used
  - to determine the name of the remote directory
- To determine the set of files available in the current directory, a user can enter the “ls” command

34.12 File Types And Transfer Modes (1)

- FTP defines two basic types of transfer that accommodate most files:
  - textual and binary
- The user must select a transfer type
  - and the mode stays in effect for the entire file transfer
- Textual transfer is used for basic text files
  - A text file contains a sequence of characters separated into lines
  - most computer systems use either the ASCII or EBCDIC
- A user, who knows a remote computer uses the EBCDIC character set can enter the command type “ebcdic”
  - to specify textual transfer and request FTP to translate between EBCDIC and the local character set when copying a file

34.12 File Types And Transfer Modes (2)

- Alternative to text transfer in FTP is “binary” transfer
  - To be used for all non-text files
  - For example, an audio clip, a graphics image, or a matrix of floating-point numbers must be transferred in binary mode
- A user enters the “binary” command to place FTP in binary mode
  - FTP does not interpret the contents of a file transferred in binary mode
  - and does not translate from one representation to another
  - Instead, binary transfer merely produces a copy
  - the bits of a file are reproduced without change
- A binary transfer may not produce the expected result

34.14 Verbose Output

- output from FTP begins with a three-digit number that identifies the message.
- FTP places 226 at the beginning of a message that informs the user that a transfer has completed.
- Remote FTP places 221 at the beginning of a message that ACKs a request to close the control connection
- A user can choose
  - an FTP should issue informational messages "verbose mode"
  - or omit such messages and report only the results "quiet mode"
- To control the mode, a user enters the "verbose"

34.15 Client-Server Interaction In FTP

- FTP uses the client-server paradigm
  - A user runs a local FTP application
  - When a user enters an "open" command and specifies a remote host
    - the local application becomes an FTP client that uses TCP to establish a control connection to an FTP server on the specified computer
- The client/server use the FTP protocol to communicate
  - That is, the client does not pass the user's keystrokes directly to the server
  - Instead, when a user enters a command, the client interprets the command
    - If the command requires interaction with the server, the client forms a request using the FTP protocol, and sends the request to the server

34.16 Control And Data Connections

- FTP uses a control connection only to send commands and receive responses
- When it transfers a file, FTP does not send the data across the control connection
  - Instead, the client and server establish a separate data connection for each file transfer, use it to send one file, and then close the connection
  - If the user requests another transfer
    - the client and server establish a new data connection
- To avoid conflict between the control and data connections
  - FTP uses a different protocol port number for each of them
  - Although data connections appear and disappear frequently
    - the control connection persists for the entire session
- Figure 34.2 illustrates the concept

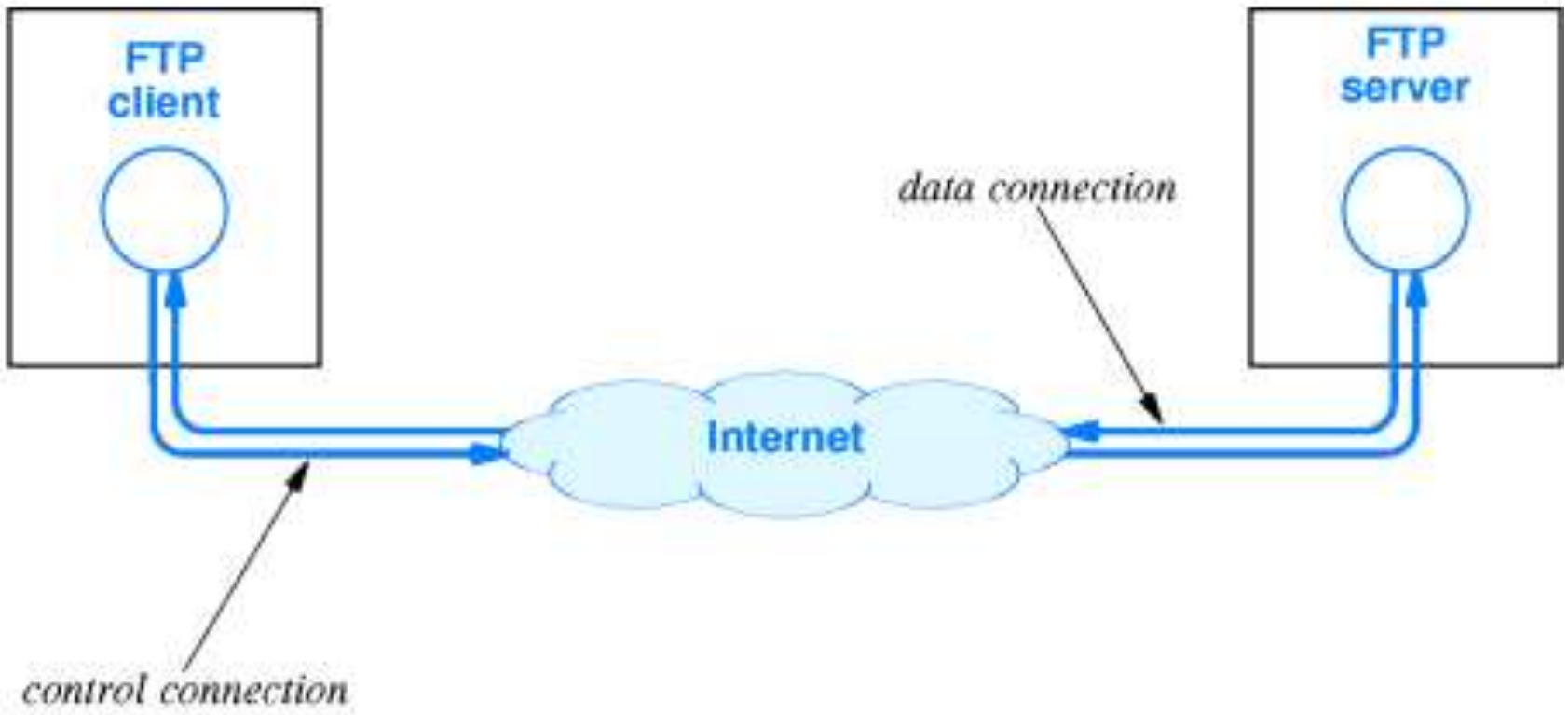

Figure 34.2 TCP connections between an FTP client and server during a file transfer, with an arrow showing the direction in which each connection was formed. Although the control connection remains open, the data connection is closed after the transfer completes.

34.17 Data Connections And End Of File

- Using separate connections for transfer and control has several advantages
  - the scheme keeps the protocols simpler and makes implementation easier → data from a file is never confused with FTP commands
  - it can be used during the transfer, because it stays in place
- EOF is used to indicate that a transfer has completed
- The server does not need to tell the client the file size in advance
  - When the server reaches the end of the file, the server closes the data connection, causing the client to receive an end-of-file condition

34.18 Trivial File Transfer Protocol (1)

Trivial File Transfer Protocol (TFTP) differs from FTP in several ways

- TFTP client/server uses UDP instead of TCP
- TFTP only supports file transfer
  - TFTP does not support interaction and does not have a large set of commands
- TFTP does not permit a user to list the contents of a directory
  - or interrogate the server to determine available files list
- TFTP does not have authorization
  - A client does not send a login name or password; a file can be transferred only if the file permissions allow global access

34.18 Trivial File Transfer Protocol (2)

- Although TFTP is less powerful than FTP, TFTP has two advantages
  - TFTP can be used when UDP is available, but TCP is not
  - TFTP requires less memory than the code for FTP
- These advantages are not important in a PC, but
  - they can be important in a small computer or a special-purpose HW device

34.18 Trivial File Transfer Protocol (3)

- TFTP is especially useful for bootstrapping
  - For devices with limited storage
  - All the device needs is a network connection and a small amount of ROM into which TFTP, UDP, and IP are hardwired
  - When it receives power, the device executes the code in ROM, which broadcasts a TFTP request across the network.
  - A TFTP server on the network is configured to answer the request by sending a file that contains the program to be run
  - The device receives the file, loads it into memory, and begins to execute the program
- Bootstrapping over a network adds flexibility and reduces cost

34.19 Network File System (1)

- To accommodate applications that only need to read or write part of a file, TCP/IP includes a file access service
- Unlike a file transfer service
  - a file access service allows a remote client to copy or change small pieces without copying an entire file
- The file access mechanism used with TCP/IP
  - is known as the Network File System (NFS)
- NFS allows an application
  - to open a remote file
  - move to a specified position in the file
  - and read or write data starting at that position

34.19 Network File System (2)

- NFS allows shared file access
  - A file that resides at an NFS server can be accessed by multiple clients
- To prevent others from interfering with file updates
  - NFS allows a client to lock a file
  - When the client finishes making changes, the client unlocks the file, allowing others access